

Physics 110 Course Policies and Syllabus

U.S. Air Force Academy, Department of Physics and Meteorology

I. COURSE DESCRIPTION

Physics 110. *General Physics I with Laboratory*. Introductory calculus-based physics course with emphasis on contemporary applications (Part I). Topics include kinematics, Newtonian mechanics (statics and dynamics), work, conservation of energy, conservation of momentum, rotational motion, Universal Law of Gravitation, orbital mechanics, simple harmonic motion, and wave motion. Possible additional topics include astronomy, thermodynamics, and relativity. Course emphasizes the use of vectors and calculus in problem solving. Course includes in-class laboratories and computer applications to highlight key concepts. Final exam. Coreq: Math 142.

II. COURSE PREREQUISITES

You must have completed (or be currently enrolled in) Math 142. You are expected to be knowledgeable and proficient in the following math concepts/skills:

- algebra and trigonometry
- vector operations, including dot product and cross product
- differentiation of polynomials and simple functions

III. COURSE GOALS

Core physics courses, including Physics 110 and Physics 215, contribute to the development and assessment of the *Scientific Reasoning and the Principles of Science* outcome, to apply scientific habits of mind, including proficiency in the nature of science, scientific reasoning, and the principles of science. This includes the following general proficiencies:

1. *Nature of Science* – Determine if explanations, claims, or statements are consistent with the Nature of Science and explain how a scientific explanation can simultaneously be broadly accepted as reliable yet provisional.
2. *Scientific Reasoning* – Use if-then reasoning to generate a scientifically testable expectation; describe the effects of data collection methods on the reliability of results; use mathematics and appropriate technology to describe and represent data (e.g., symbolically, visually, or numerically); and systematically interpret results to draw valid conclusions.
3. *Principles of Science* – Identify, explain, and discuss the role of foundational science principles in real-world applications and apply foundational principles of science and mathematics to predict or explain natural phenomenon

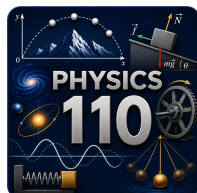
Additionally, during Physics 110 you will learn to:

1. Describe the motion of objects using kinematics.
2. Interpret and solve motion problems using Newton's three laws.
3. Analyze the motion of objects using conservation of energy, momentum, and angular momentum.
4. Develop valid physics-based conclusions about real-world problems and applications.

IV. COURSE ADMINISTRATION

Head, Department of Physics and Meteorology: Col Rose Tseng
 Director of Core Physics: LTC Tyler Jones
 Course Director: LTC Tyler Jones

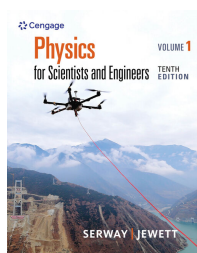
V. COURSE MATERIAL



Physics 110 Team

Each cadet is required to purchase website access to Cengage WebAssign, which comes with access to an eBook version of the textbook. Students are encouraged, but not required, to purchase a hard copy of the text. Your Cengage WebAssign account will be linked to this course through your blackboard account. Course announcements will be sent through the “Physics 110 Fall 2026” team in Microsoft Teams. Any other required materials (i.e., lab handouts, supplemental worksheets) will be available in the ‘Shared’ folder of the Teams channel.

- **WebAssign** (REQUIRED) eBook, assignments, and additional resources (worked examples, instructional videos, etc.) through Cengage.
 - Purchase **WebAssign Cengage Unlimited**
- **Textbook** (ENCOURAGED) R. A. Serway, J. W. Jewett, *Physics for Scientists and Engineers with Modern Physics, 10th Edition* (Cengage Learning, 2019).
 - Hardcover ISBN-13: 9781337553292



Physics for Scientists and Engineers with Modern Physics

VI. GRADED EVENTS

Graded Event	Number	Points (Each)	Total (%)
Preflight	37	2	74 (7%)
Homework	28	3	84 (8%)
Labs	6	20	120 (11.5%)
Exam-Prep Quiz	6	20	120 (11.5%)
Cumulative Homework Set	4	10	40 (4%)
Graded Reviews	3	100	300 (29%)
Final Exam	1	300	300 (29%)
Total			1038

Important Notes about Course Grades:

- Failure to score more than 40% on the final exam does not demonstrate basic understanding of the course learning objectives, and may result in an automatic failure of the course. Other factors that may be considered include: performance on prior GRs and EPQs.
- The gradelines at Prog and the end of the course are not fixed. They are determined by course and department leadership.
- Successful completion of the preflight and homework assignments is vital for your learning.

VII. COURSE WORK

Most lessons in this course include two out-of-class assignments: pre-class preparation in the form of a short reading and preflight assignment, and post-class homework. You should expect to spend approximately an hour on each of these components. It is important that you put meaningful effort into these tasks. If you are spending significantly more time than this for each lesson you should seek EI from your instructor.

Successful completion of the preflight and homework assignments, including cumulative homework sets, is vital for your learning and accounts for approximately 19% of your overall grade. If you earn an average of 80% on your labs, EPQs, and GRs, but only complete 50% of your preflights and homework, you will go into the final exam with a 74% in the course. If you instead earn 95% on your preflights and homework, you will go into the final exam with an 83% in the course. Giving up these points by missing assignments will have a direct negative impact on your grade, amplified because less effort spent working through these assignments will likely lead to lower EPQ and GR scores.

A. Reading

There will be a reading assignment to complete every lesson as noted on the syllabus.

B. Preflight

With the exception of Lesson 1 and GRs, every lesson will have a preflight assignment to be completed/submitted by 2359 the night before the lesson is taught to your section. The Pre-lesson Readiness Engagement Platform (PREP) and Interactive Pre-lesson Readiness Engagement Platform (iPREP) will serve as the preflight platforms for this course. PREP allows you to answer short conceptual questions about the assigned pre-class materials, while iPREP engages you in an interactive dialogue about the lesson content. These assignments are designed to introduce key concepts, assess your understanding of the assigned material, and provide feedback to your instructor before class. You will receive credit based on participation, effort, scientific reasoning, and critical thinking rather than solely on the correctness of your answers. Links to PREP and iPREP will be available through the course Blackboard site. Both platforms are entirely web-based and do not require you to install any software. Using iPREP requires either a free or paid Claude account. Unless otherwise specified, preflight assignments are due by 2359 on the night before the lesson is taught to your section. Each preflight assignment is worth 2 points. If excused from class for an approved SCA absence, any pre-class work must be done prior to the absence. You should make every effort to understand the course material before coming to class. Class time will be spent answering questions, honing your knowledge, practicing problems, and correcting misconceptions.

C. Homework

With the exceptions annotated on the schedule, every lesson will have a homework assignment. Homework is your third exposure to the lesson material and is meant to solidify conceptual understanding and problem-solving skills. Homework for a specific lesson will become available prior to that lesson and must be completed via WebAssign before 2359 the night before the second lesson after the homework is assigned. For example, problems assigned for Lesson 5 material are due by 2359 the night before lesson 7. This allows time for you to ask your instructor clarifying homework questions the following lesson. Each homework set is worth 3 points. Each question will have 5 attempts.

Additionally, before each graded review and the final there is a cumulative homework set, indicated by a **C** on the syllabus. This problem set serves as a refresher for material presented throughout the block, but will not cover all testable material (i.e., this should *not* be your only GR or final study). Each review set is worth 10 points and is due at 2359 before each GR.

D. Math Skills Enhancement (MSE)

Math Skills Enhancement (MSE) exercises will be available periodically throughout the semester to practice foundational math skills which are critical to your success in Physics 110. MSE's are optional but highly encouraged activities that should contribute to your success. Successful completion of these exercises can earn you up to 20 course

points back for any points you miss on preflight or homework assignments. MSE's are completed in WebAssign and are due on Sunday evenings by 2359 for the following dates: Aug 9, Aug 23, Aug 30, Sept 6, Sept 20, Sept 27, Oct 4, Oct 11, Oct 25, and Nov 1 as shown in WebAssign.

E. Labs

On lab days, classes will span the full double-period time (113 min). During this time, your lab group will apply experimental skills to investigate physics principles and complete data collection and analysis. Each group will prepare a single lab report and submit it in GradeScope. **Lab reports will typically be due at 1159 the lesson following the lab** (specific due dates will be published and available in Gradescope). Each lab is worth 20 points. At the Course Directors discretion, some labs may have the option to earn up to 2 bonus points by successfully completing additional tasks stated in the lab prompt, however, the maximum lab score is capped at 20 points per lab.

F. Double Periods

PHYS 110 is a 4.0 semester hour course. Each class will be a double-period, and you can expect a typical lesson (not including labs) to conclude after 90 minutes, providing time for you to ask any questions you have or get a start on your homework assignment.

G. Exam-Prep Quizzes

Exam-Prep Quizzes (EPQs) consist of multiple-choice and workout problems like those on GRs and the final exam. They will be administered during class time, and you will have 25 minutes to complete each EPQ. You should use these quizzes to gauge your understanding of the material before the exams, along with assess the laboratory learning objectives for this course. There will be 6 EPQs, each worth 20 points. These are individual-effort assessments, and collaboration during exams or quizzes are strictly forbidden. Communication with another cadet during an exam or quiz will result in a grade of zero.

H. Graded Reviews

Graded Reviews (GRs) typically consist of multiple-choice questions, short answer questions, and workout problems. They will be administered during the common GR period, and you will have 50 minutes to complete each GR. There will be 3 GRs, each worth 100 points. These are individual-effort assessments, and collaboration during exams or quizzes are strictly forbidden. Communication with another cadet during an exam or quiz will result in a grade of zero.

I. Final Exam

There will be a comprehensive final exam at the end of the semester. The final exam is your opportunity to demonstrate your mastery of the material. As such, validation of the final exam is not offered. The final exam is worth 300 points.

VIII. DOCUMENTATION POLICY AND UNAUTHORIZED RESOURCES

DFPM's documentation standards align with DF policy. DFPM encourages collaboration and group work to help you master challenging concepts, and department policies regarding unauthorized resources are focused on eliminating shortcuts and workarounds that have been shown to hinder the learning process. In general, **unauthorized resources** are resources that do not help you learn, but rather provide an answer to a problem without developing the understanding of how to solve it. **Unauthorized resources should not be used when completing assignments for a grade.** One example:
- **Authorized resource:** Asking your roommate for assistance solving a homework

problem because you have attempted it and were confused on a step.

- **Unauthorized resource:** Asking your roommate for the numerical solution to a homework problem.

Solution-providing websites such as Chegg.com are not authorized resources for core physics courses. These websites are not effective learning tools and can too easily circumvent the learning that takes place when you actually work through an assigned problem.

Generative AI or solution providing website usage guidance. Based on the USAFA GenAI guidance memo, this course allows level 3 usage of GenAI, including use on lab assignments. AI platforms and solution providing websites can be either authorized or unauthorized resources, depending on how you interact with them. Using GenAI to guide you through the physics of a homework problem, much like you would use a textbook, is authorized. Among the benefits of AI is the ability to generate an explanation of a concept and then ask follow-on questions to define terms or expand on parts of the explanation. Using these tools in this fashion will generally enhance learning and DFPD encourages exploration with this technology. Looking at a generated solution or directly asking an AI platform for a solution to a problem on a graded assignment (just like asking someone to do your homework for you) is unauthorized. Technology used in this way is not effective for learning and can too easily circumvent the learning that takes place when you actually work through an assigned problem.

EPQs and GRs are **individual effort**; all other assignments are **collaborative**. You do not need to document use of your course textbook, class slides/handouts, or assistance you receive from your instructor, but always document all other resources you reference as you complete graded assignments. Regardless of what resources you use, or how you use them, accurately describe the level of assistance received. Assignments completed using an unauthorized resource will not receive academic credit, but proper documentation is essential to avoid any concerns about honor or academic integrity. If you have any questions about authorized or unauthorized resources, please discuss with your course instructor.

Academic Security: You should be cognizant of the academic security instructions indicated on each graded event. Some graded events are never released from academic security.

IX. ABSENCES, TARDINESS, AND MAKE-UP EXAMS

If you will miss a lesson for any reason, you will still be expected to complete all preflight and homework assignments on the due date, unless you have been granted an extension by your instructor, otherwise no credit will be given. Only absences approved by the SCA committee or authorized by USAFA instruction (e.g., SqCC authorized bedrest) will be considered excused.

Squadron Commander Bedrest

Students on Commander bedrest must e-mail their instructor AND copy their Squadron Commander and AMT *before* missing class to show that their bedrest is approved. Otherwise, the student will be marked absent and any missed work will be subject to a 25% penalty.

Labs

If you will be absent or more than 20 minutes late, you will be required to complete your own lab or join a makeup group. For excused absences, you must complete missed labs within 3 lessons (6 business days) of the originally scheduled date. You alone are

responsible for scheduling a time to make up any missed in-class activity, regardless of the reason. Exceptions to this policy must be approved by the course director.

Graded Reviews

If you will be absent during a GR due to a Scheduling Committee Action (SCA), you are responsible to notify your instructor at least **THREE BUSINESS DAYS** (i.e., not including weekends) **PRIOR** to the offering of the exam. If you are more than 15 minutes late unexcused for a GR, you must take a makeup exam with a 25% penalty. If you are less than 15 minutes late, you may still take the exam during the remaining scheduled time. Work with your instructor to schedule a time to make up the exam or quiz within three lessons.

Make-up Exams (GRs and EPQs)

Make-up exams will only be offered without penalty for students absent with an excused absence. Exams will not be administered on the road unless approved by the Course Director on a case-by-case basis. Work with your instructor to schedule a time to make up the exam or quiz within three lessons.

Cut-Off Work Date

For the semester, all late/make-up work is due by COB (1700) T41, any assignment not turned in at that time will receive a zero unless there is a mitigating circumstance (injury/illness) which prevents completion of the assignment and has been shared with your instructor before the T41 due date. **The last Homework assignment is due on 11 December 2026.**

X. EXTRA INSTRUCTION (EI)

You are highly encouraged to seek Extra Instruction (EI) as frequently as needed. Experience shows that students who routinely access EI, who come prepared to each EI session, and who take ownership of their learning do well in challenging courses, such as PHYS 110. Please coordinate with your instructor for either virtual or in-person EI sessions. You may find that the Quantitative Reasoning Center is a useful resource available after hours to provide EI. Appointments can be made online, and by joining their Teams page you can find additional resources and practice sets that may prove beneficial to your learning of course material.

XI. LEARNING OBJECTIVES

The graded events in Physics 110 are designed to assess your mastery of a variety of learning objectives. These objectives are divided into four major blocks of course material, one block of lab skills, and one block of USAFA learning outcomes.

Block 1 Objectives: Kinematics and Newton's Laws

- Obj 1-1** Develop and apply problem solving strategies to solve scientific problems involving an object's motion.
- Obj 1-2** Use the calculus-based relationships between position, velocity, and acceleration to analyze an object's motion in one and two dimensions.
- Obj 1-3** Given a graph of position, velocity, or acceleration of a moving object, calculate the related variables, including instantaneous and average values.
- Obj 1-4** Use appropriate equations of motion and vector mathematics to calculate the trajectory of a projectile.
- Obj 1-5** Depict, using a free-body diagram, realistic forces on an object, including gravitational, normal, tension, friction, drag, spring, applied, and thrust.
- Obj 1-6** Calculate, using appropriate vector notation and mathematics, the net force on an object.
- Obj 1-7** Define uniform circular motion, explain why uniform circular motion involves acceleration, and describe the direction of that acceleration.
- Obj 1-8** Calculate the velocity and acceleration of an object whose motion is uniformly circular.
- Obj 1-9** Use Newton's laws of motion to solve problems involving multiple forces and/or multiple objects, to include calculating linear and/or centripetal acceleration.

Block 2 Objectives: Energy and Linear Momentum

- Obj 2-1** Using appropriate calculus and vector mathematics, calculate the work done by constant forces and by forces that are functions of position.
- Obj 2-2** Calculate the kinetic energy of a system and use the work-kinetic energy theorem to calculate the change in a system's kinetic energy.
- Obj 2-3** Calculate the rate at which energy is transferred using power.
- Obj 2-4** Solve problems, including work done by conservative, non-conservative, and/or applied forces.
- Obj 2-5** Evaluate the potential energy associated with systems that involve different conservative forces.
- Obj 2-6** Use linear momentum to calculate the impulse an object experiences.
- Obj 2-7** Apply the conservation of linear momentum and conservation of kinetic energy, when applicable, to solve problems involving elastic or inelastic collisions or problems involving single object break-apart events (e.g., explosions).
- Obj 2-8** Calculate the center of mass for systems of discrete masses.

Block 3 Objectives: Rotational Motion and Gravitational Forces

- Obj 3-1** Use the angular analogs to the linear equations of motion to solve problems that involve angular displacement, angular velocity, and angular acceleration.
- Obj 3-2** Calculate, using appropriate vector notation and mathematics, the torque on an object.
- Obj 3-3** Determine the moment of inertia for a system of discrete particles, rigid objects, or a combination of both.
- Obj 3-4** Use Newton's Second Law and its rotational analog to solve problems involving translational motion and/or rotational motion.
- Obj 3-5** Solve problems involving rotating objects using rotational kinetic energy and conservation of energy.
- Obj 3-6** Determine the angular momentum for discrete particles and rotating rigid objects and apply conservation of angular momentum to solve problems that involve rotating objects and collisions.
- Obj 3-7** Using universal gravitation, solve problems involving the gravitational interaction between two objects.
- Obj 3-8** Using universal gravitation, calculate the speed, acceleration, mechanical energy, and period of an object in a circular orbit.
- Obj 3-9** Use the concept of conservation of mechanical energy to calculate the energy characteristics of open and closed orbits.

Block 4 Objectives: Simple Harmonic Oscillators and Wave Motion

- Obj 4-1** Derive the equations of motion of spring and pendulum simple harmonic oscillators from their physical parameters and analyze mechanical energy of the system.
- Obj 4-2** Describe and compare underdamped, critically damped, and overdamped oscillatory motion, and predict how damping causes the amplitude and mechanical energy of an oscillator to decrease over time.
- Obj 4-3** Describe and analyze traveling mechanical waves in terms of amplitude, wavelength, frequency, wave number, angular frequency, period, phase, and propagation speed, and apply the wave equation to solve quantitative problems.
- Obj 4-4** Use the principle of superposition to determine how waves combine in the same region of space to form interference patterns and standing waves..

Lab Objectives

- Obj L-1** Develop confidence in the scientific method as an approach to investigate everyday phenomenon.
- Obj L-2** Develop symbolic expressions that accurately depict a theoretical model being examined by an experiment.
- Obj L-3** Develop lab and data collection techniques suitable for analysis and reporting results.
- Obj L-4** Analyze the quantitative results of an experiment and develop appropriate conclusions about the physical world using mathematical techniques

Obj L-5 Understand the impact of measurement uncertainty and error in experimentation and develop experimental and statistical mitigation techniques.

Obj L-6 Communicate scientific reasoning and experimental data through structured, clear technical writing and appropriate visual representations.

USAFA Outcome Objectives

Obj-1 Determine if explanations or claims are consistent with the Nature of Science. (Proficiency 1)

Obj-2 Evaluate the validity of study design and data collection methods in scientific investigations. (Proficiency 4)

Obj-3 Represent and interpret data using mathematics and graphics. (Proficiency 5)

Obj-4 Explain phenomenon using fundamental scientific principles that apply across the natural sciences. (Proficiency 7)

TABLE 1. PHYS 110 Fall 2026 Course Schedule

Block	LSN	M-Day	T-Day	Topic	Reading	In-Class	Pre-Fit	HW
Kinematics	1	Thu, Aug 6	Fri, Aug 7	Course Admin / Physics and Measurement	1.1-1.6, Appdx B	Baseline Quiz; DP 1	N	Y
	2	Mon, Aug 10	Tue, Aug 11	1-D Motion / Position Velocity Acceleration	2.1-2.6		Y	Y
	3	Wed, Aug 12	Thu, Aug 13	Constant Acceleration and Freefall	2.7-2.8		Y	Y
	4	Fri, Aug 14	Mon, Aug 17	Vectors and Coordinate Systems	3.1-3.4		Y	Y
	5	Tue, Aug 18	Wed, Aug 19	2-D Motion / Projectile Motion	4.1-4.3		Y	Y
	6	Thu, Aug 20	Fri, Aug 21	<i>Laboratory Fundamentals</i>	Lab Manual	EPQ 1; DP 2	Y	N
	7	Mon, Aug 24	Tue, Aug 25	<i>LAB 1: Projectile Motion</i>	Lab Handout	LAB 1	Y	N
Newton's Laws	8	Wed, Aug 26	Thu, Aug 27	Intro to Newton's Laws	5.1-5.7		Y	Y
	9	Fri, Aug 28	Mon, Aug 31	Application of Newton's Laws / Friction	5.6-5.8		Y	Y
	10	Tue, Sept 1	Wed, Sept 2	<i>LAB 2: Newton's Laws</i>	Lab Handout	LAB 2	Y	N
	11	Thu, Sept 3	Fri, Sept 4	Newton's Laws with Circular Motion	6.1-6.2; 4.4-4.5	EPQ 2; DP 3	Y	Y
	12	Tue, Sept 8	Wed, Sept 9	Application of Newton's Laws	5.1-6.3		Y	C
	13	Mon, Sept 14	Graded Review 1			GR 1	N	N
Energy	14	Tue, Sept 15	Wed, Sept 16	Energy of a System / Work	7.1-7.3, 8.1		Y	Y
	15	Thu, Sept 17	Mon, Sept 21	Varying Forces / Kinetic Energy	7.4-7.5		Y	Y
	16	Tue, Sept 22	Wed, Sept 23	Potential Energy / Nonconservative Forces	7.6-7.9		Y	Y
	17	Thu, Sept 24	Fri, Sept 25	Conservation of Energy	8.1-8.2	DP 4	Y	Y
	18	Mon, Sept 28	Tue, Sept 29	Changes in Mechanical Energy / Power	8.3-8.5	EPQ 3	Y	Y
	19	Wed, Sept 30	Thu, Oct 1	<i>LAB 3: Conservation of Energy</i>	Lab Handout	LAB 3	Y	N
Momentum	20	Fri, Oct 2	Mon, Oct 5	Linear Momentum & 1-D Collisions	9.1-9.4		Y	Y
	21	Tue, Oct 6	Wed, Oct 7	2-D Collisions	9.5	DP 5	Y	Y
	22	Thu, Oct 8	Fri, Oct 9	Center of Mass / Systems of Particles	9.6-9.7	EPQ 4	Y	Y
	23	Tue, Oct 13	Wed, Oct 14	<i>LAB 4: Conservation of Momentum</i>	Lab Handout	Lab 4	Y	C
	24	Thu, Oct 15	Graded Review 2			GR 2	N	N
Rotating Objects	25	Mon, Oct 19	Tue, Oct 20	Angular Position Velocity Acceleration	10.1-10.3; 4.4-4.5		Y	Y
	26	Wed, Oct 21	Thu, Oct 22	Vector Product / Torque / Moment of Inertia	10.4-10.5 / 11.1		Y	Y
	27	Fri, Oct 23	Mon, Oct 26	Moment of Inertia Applications	10.5	DP 6	Y	Y
	28	Tue, Oct 27	Wed, Oct 28	Rotational Kinetic Energy / Rolling Motion	10.7, 10.9		Y	Y
	29	Thu, Oct 29	Fri, Oct 30	Angular Momentum	11.2-11.3		Y	Y
	30	Mon, Nov 2	Tue, Nov 3	Conservation of Angular Momentum	11.4	EPQ 5	Y	Y
	31	Wed, Nov 4	Thu, Nov 5	Rotational Motion Applications	11.4	DP 7	Y	Y
	32	Fri, Nov 6	Mon, Nov 9	<i>LAB 5: Angular Momentum</i>	Lab Handout	Lab 5	Y	N
Gravity	33	Tue, Nov 10	Thu, Nov 12	Universal Gravitation	13.1-13.3	Baseline Quiz	Y	Y
	34	Fri, Nov 13	Mon, Nov 16	Gravitational Potential Energy / Kepler's Laws	13.4-13.5		Y	Y
	35	Tue, Nov 17	Wed, Nov 18	Planetary and Satellite Motion	13.6		Y	C
	36	Thu, Nov 19	Graded Review 3			GR 3	N	N
Oscillatory Motion	37	Mon, Nov 23	Tue, Nov 24	Simple Harmonic Motion & Pendulums	15.1-15.2, 15.5		Y	Y
	38	Tue, Dec 1	Wed, Dec 2	<i>Lab 6: Simple Harmonic Motion</i>	Lab Handout	LAB 6	Y	N
	39	Thu, Dec 3	Fri, Dec 4	Energy of SHO & Damped Oscillations	15.3, 15.6		Y	Y
	40	Mon, Dec 7	Tue, Dec 8	Wave Fundamentals & Wave Equation	16.1-16.3	EPQ 6	Y	Y
	41	Wed, Dec 9	Thu, Dec 10	Superposition, Standing Waves, & Applications	17.1-17.2		Y	C

- "Y" (Yes) and "N" (No) designate if there is a Preflight or Homework for the corresponding lesson
- In-class Discovery Project (DP), a hands-on in-class exercise